AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 20: Gaussian Integrals

Adam Prugel-Bennett

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1 Keywords

- Gaussian integrals, completing the square, moment generating functions, marginals, conditionals, Kalman filter

2 Main Points

2.1 Why Gaussians?

- Gaussians are used throughout machine learning (Gaussian processes, Kalman filters, linear regression, the noise in diffusion models) because all their integrals can be done exactly
- Marginalising, conditioning, adding or linearly transforming Gaussians gives another Gaussian, so inference reduces to linear algebra on means and covariances
- The algebra is long-winded but uses only three tricks: the basic Gaussian integral, completing the square, and diagonalising a matrix. Learn the results; read the derivations once so you trust them

2.2 One Dimension

- The basic integral, by squaring it and changing to polar coordinates,

$$\int_{-\infty}^{\infty} \mathrm{e}^{-x^2/2}\,\mathrm{d}x = \sqrt{2\pi}$$

  and, changing variables to $z = (x-\mu)/\sigma$, the normal density

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\,\sigma}\,\mathrm{e}^{-(x-\mu)^2/(2\sigma^2)}$$

  integrates to one

- **Completing the square** (for $a > 0$): $-\frac{a}{2}x^2 + bx = -\frac{a}{2}(x - b/a)^2 + b^2/(2a)$, so

$$\int_{-\infty}^{\infty} \mathrm{e}^{-\frac{a}{2}x^2 + b\,x}\,\mathrm{d}x = \sqrt{\frac{2\pi}{a}}\,\mathrm{e}^{\frac{b^2}{2a}}$$

- **Moment generating function** (MGF) $M(t) = \mathbb{E}\left[\mathrm{e}^{tX}\right]$. For $X \sim \mathcal{N}(\mu, \sigma^2)$

$$M(t) = \mathrm{e}^{\mu t + \frac{1}{2}\sigma^2 t^2}$$

  - $M^{(n)}(0) = \mathbb{E}[X^n]$, so $M'(0) = \mu$ and $M''(0) = \mu^2 + \sigma^2$
  - the cumulant generating function $\log M(t)$ stops at t^2: a Gaussian has no cumulants beyond the mean and variance

- An MGF determines its distribution, and for independent X, Y the MGF of $X + Y$ is the product of their MGFs. So the sum of independent Gaussians is Gaussian, with the means added and the variances added

2.3 Many Dimensions

- For a positive definite covariance $\mathbf{\Sigma} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\mathsf{T}$, rotating to $\boldsymbol{y} = \mathbf{V}^\mathsf{T}(\boldsymbol{x} - \boldsymbol{\mu})$ (Jacobian $\mathbf{V}$, with $|\det \mathbf{V}| = 1$) turns the integral into a product of one-dimensional ones:

$$\int \mathrm{e}^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\mathsf{T}\mathbf{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})}\,\mathrm{d}\boldsymbol{x} = \prod_i \sqrt{2\pi\lambda_i} = \sqrt{\det(2\pi\mathbf{\Sigma})}$$

  so

$$\mathcal{N}(\boldsymbol{x} \mid \boldsymbol{\mu}, \mathbf{\Sigma}) = \frac{1}{\sqrt{\det(2\pi\mathbf{\Sigma})}}\,\mathrm{e}^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\mathsf{T}\mathbf{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})}$$

- **The central result.** Completing the square, $-\frac{1}{2}\boldsymbol{x}^\mathsf{T}\mathbf{A}\boldsymbol{x} + \boldsymbol{b}^\mathsf{T}\boldsymbol{x} = -\frac{1}{2}(\boldsymbol{x} - \mathbf{A}^{-1}\boldsymbol{b})^\mathsf{T}\mathbf{A}(\boldsymbol{x} - \mathbf{A}^{-1}\boldsymbol{b}) + \frac{1}{2}\boldsymbol{b}^\mathsf{T}\mathbf{A}^{-1}\boldsymbol{b}$, gives

$$\int \mathrm{e}^{-\frac{1}{2}\boldsymbol{x}^\mathsf{T}\mathbf{A}\boldsymbol{x} + \boldsymbol{b}^\mathsf{T}\boldsymbol{x}}\,\mathrm{d}\boldsymbol{x} = \sqrt{\frac{(2\pi)^n}{\det \mathbf{A}}}\,\mathrm{e}^{\frac{1}{2}\boldsymbol{b}^\mathsf{T}\mathbf{A}^{-1}\boldsymbol{b}}$$

  (sometimes called the Hubbard–Stratonovich transformation)

- The MGF of $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{\Sigma})$ is

$$\mathbb{E}\left[\mathrm{e}^{\boldsymbol{t}^\mathsf{T}\boldsymbol{x}}\right] = \mathrm{e}^{\boldsymbol{t}^\mathsf{T}\boldsymbol{\mu} + \frac{1}{2}\boldsymbol{t}^\mathsf{T}\mathbf{\Sigma}\,\boldsymbol{t}}$$

  from which $\mathbb{E}[\boldsymbol{x}] = \boldsymbol{\mu}$ and the covariance is $\mathbf{\Sigma}$

- **Linear transformations.** If $\boldsymbol{z} = \mathbf{W}\boldsymbol{x} + \boldsymbol{c}$ then $\mathbb{E}\left[\mathrm{e}^{\boldsymbol{t}^\mathsf{T}\boldsymbol{z}}\right] = \mathrm{e}^{\boldsymbol{t}^\mathsf{T}\boldsymbol{c}}\,M(\mathbf{W}^\mathsf{T}\boldsymbol{t})$, so

$$\boldsymbol{z} \sim \mathcal{N}\big(\mathbf{W}\boldsymbol{\mu} + \boldsymbol{c},\ \mathbf{W}\mathbf{\Sigma}\mathbf{W}^\mathsf{T}\big)$$

  ($\mathbf{W}$ need not be square)

2.4 Marginals and Conditionals

- Partition a Gaussian vector into $\boldsymbol{z}_1$ and $\boldsymbol{z}_2$, with means $\boldsymbol{\mu}_1$, $\boldsymbol{\mu}_2$ and covariance blocks $\mathbf{\Sigma}_{11}$, $\mathbf{\Sigma}_{12}$, $\mathbf{\Sigma}_{21} = \mathbf{\Sigma}_{12}^\mathsf{T}$, $\mathbf{\Sigma}_{22}$
- **Marginal.** Setting $\boldsymbol{t} = (\boldsymbol{t}_1, \mathbf{0})$ in the MGF gives the MGF of $\mathcal{N}(\boldsymbol{\mu}_1, \mathbf{\Sigma}_{11})$: to marginalise, strike out the rows and columns of the variables integrated away
- **Conditional.** Let $\boldsymbol{w} = \boldsymbol{z}_1 - \mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1}\boldsymbol{z}_2$. Then $\mathrm{Cov}(\boldsymbol{w}, \boldsymbol{z}_2) = \mathbf{\Sigma}_{12} - \mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1}\mathbf{\Sigma}_{22} = \mathbf{0}$. Jointly Gaussian variables that are uncorrelated are independent (their joint MGF factorises), so $\boldsymbol{w}$ is independent of $\boldsymbol{z}_2$, and

$$\boldsymbol{z}_1 \mid \boldsymbol{z}_2 \sim \mathcal{N}\big(\boldsymbol{\mu}_1 + \mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1}(\boldsymbol{z}_2 - \boldsymbol{\mu}_2),\ \mathbf{\Sigma}_{11} - \mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1}\mathbf{\Sigma}_{21}\big)$$

  - the mean moves linearly with the observation; the covariance shrinks by an amount that does not depend on the value observed
  - $\mathbf{\Sigma}_{11} - \mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1}\mathbf{\Sigma}_{21}$ is called the *Schur complement* of $\mathbf{\Sigma}_{22}$
- **The long way round.** The same results can be obtained by doing the integrals directly; see the derivation below

2.5 Inference with Gaussians

- **A noisy measurement.** Prior $x \sim \mathcal{N}(\mu_0, s_0^2)$, measurement $y = x + \epsilon$ with $\epsilon \sim \mathcal{N}(0, \sigma^2)$. Then (x, y) is jointly Gaussian with covariance $\begin{pmatrix} s_0^2 & s_0^2 \\ s_0^2 & s_0^2 + \sigma^2 \end{pmatrix}$ and

$$\mathbb{E}[x \mid y] = \mu_0 + \frac{s_0^2}{s_0^2 + \sigma^2}(y - \mu_0), \qquad \frac{1}{\mathrm{Var}(x \mid y)} = \frac{1}{s_0^2} + \frac{1}{\sigma^2}$$

  precisions (inverse variances) add

- **Combining sensors.** Two measurements with noise variances σ_1^2, σ_2^2 and a flat prior give the precision-weighted average $(y_1/\sigma_1^2 + y_2/\sigma_2^2)/(1/\sigma_1^2 + 1/\sigma_2^2)$, with precision $1/\sigma_1^2 + 1/\sigma_2^2$

- **Predicting one variable from another.** For two variables with correlation ρ, $\mathbb{E}[x_2 \mid x_1] = \mu_2 + \rho\,(\sigma_2/\sigma_1)(x_1 - \mu_1)$ and $\mathrm{Var}(x_2 \mid x_1) = \sigma_2^2(1-\rho^2)$: linear regression, and regression to the mean

- **Kalman filter.** For a drifting position $x_{t+1} = x_t + \eta_t$, $\eta_t \sim \mathcal{N}(0, q)$, observed as $y_t = x_t + \epsilon_t$, the belief $\mathcal{N}(m_t, v_t)$ is updated by a *predict* step (a sum of Gaussians: the variance becomes $v_t + q$) and an *update* step (a noisy measurement):

$$k = \frac{v_t + q}{v_t + q + \sigma^2}, \qquad m_{t+1} = m_t + k\,(y_{t+1} - m_t), \qquad v_{t+1} = (1-k)(v_t + q)$$

- **Gaussian processes.** With the unknown value f_* as $\boldsymbol{z}_1$ and noisy observations $\boldsymbol{y}$ as $\boldsymbol{z}_2$, the conditional formula gives the GP prediction

$$f_* \mid \boldsymbol{y} \sim \mathcal{N}\big(\boldsymbol{k}_*^\mathsf{T}(\mathbf{K} + \sigma^2\mathbf{I})^{-1}\boldsymbol{y},\ k_* - \boldsymbol{k}_*^\mathsf{T}(\mathbf{K} + \sigma^2\mathbf{I})^{-1}\boldsymbol{k}_*\big)$$

2.6 Derivation: Marginalising and Conditioning by Integration

- This is the long way round. Take zero means (shift the variables otherwise) and write the covariance in blocks

$$\boldsymbol{\Sigma} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\mathsf{T} & \mathbf{C} \end{pmatrix}$$

  for the variables $\boldsymbol{x}$ (first block) and $\boldsymbol{y}$ (second)

- **Block factorisation.** With $\mathbf{S} = \mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{B}^\mathsf{T}$ (multiply out to check),

$$\boldsymbol{\Sigma} = \mathbf{U}\mathbf{D}\mathbf{U}^\mathsf{T}, \qquad \mathbf{U} = \begin{pmatrix} \mathbf{I} & \mathbf{B}\mathbf{C}^{-1} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}, \qquad \mathbf{D} = \begin{pmatrix} \mathbf{S} & \mathbf{0} \\ \mathbf{0} & \mathbf{C} \end{pmatrix}$$

  The inverse of $\mathbf{U}$ just flips the sign of the off-diagonal block, so

$$\boldsymbol{\Sigma}^{-1} = (\mathbf{U}^\mathsf{T})^{-1}\mathbf{D}^{-1}\mathbf{U}^{-1} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{C}^{-1}\mathbf{B}^\mathsf{T} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{S}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{C}^{-1} \end{pmatrix} \begin{pmatrix} \mathbf{I} & -\mathbf{B}\mathbf{C}^{-1} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}$$

  and, as $\det \mathbf{U} = 1$, $\det \boldsymbol{\Sigma} = \det \mathbf{S}\ \det \mathbf{C}$

- **The exponent splits.** With $\boldsymbol{u} = \boldsymbol{x} - \mathbf{B}\mathbf{C}^{-1}\boldsymbol{y}$,

$$(\boldsymbol{x}^\mathsf{T}, \boldsymbol{y}^\mathsf{T})\,\boldsymbol{\Sigma}^{-1} \begin{pmatrix} \boldsymbol{x} \\ \boldsymbol{y} \end{pmatrix} = \boldsymbol{u}^\mathsf{T}\mathbf{S}^{-1}\boldsymbol{u} + \boldsymbol{y}^\mathsf{T}\mathbf{C}^{-1}\boldsymbol{y}$$

- **Marginal.** Integrating over $\boldsymbol{x}$ is the same as integrating over $\boldsymbol{u}$ (a shift), which gives $\sqrt{\det(2\pi\mathbf{S})}$, so

$$p(\boldsymbol{y}) = \frac{\sqrt{\det(2\pi\mathbf{S})}}{\sqrt{\det(2\pi\boldsymbol{\Sigma})}}\,\mathrm{e}^{-\frac{1}{2}\boldsymbol{y}^\mathsf{T}\mathbf{C}^{-1}\boldsymbol{y}} = \frac{1}{\sqrt{\det(2\pi\mathbf{C})}}\,\mathrm{e}^{-\frac{1}{2}\boldsymbol{y}^\mathsf{T}\mathbf{C}^{-1}\boldsymbol{y}} = \mathcal{N}(\boldsymbol{y} \mid \mathbf{0}, \mathbf{C})$$

  using $\det(2\pi\boldsymbol{\Sigma}) = \det(2\pi\mathbf{S})\ \det(2\pi\mathbf{C})$

- **Conditional.** Dividing the joint density by $p(\boldsymbol{y})$ leaves

$$p(\boldsymbol{x} \mid \boldsymbol{y}) = \frac{1}{\sqrt{\det(2\pi\mathbf{S})}}\,\mathrm{e}^{-\frac{1}{2}(\boldsymbol{x} - \mathbf{B}\mathbf{C}^{-1}\boldsymbol{y})^\mathsf{T}\mathbf{S}^{-1}(\boldsymbol{x} - \mathbf{B}\mathbf{C}^{-1}\boldsymbol{y})} = \mathcal{N}\big(\boldsymbol{x} \mid \mathbf{B}\mathbf{C}^{-1}\boldsymbol{y},\ \mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{B}^\mathsf{T}\big)$$

  the same result as the light derivation in the lecture

3 Exercises

3.1 Fourth moment

- Use the MGF of $\mathcal{N}(0, \sigma^2)$ to show $\mathbb{E}\left[X^4\right] = 3\sigma^4$. (Expand $\mathrm{e}^{\sigma^2 t^2/2}$ as a power series and compare with $\sum_n \mathbb{E}[X^n]\, t^n/n!$.)

3.2 Combining sensors

- Two sensors read $y_1 = 10.0$ (noise standard deviation 1) and $y_2 = 12.0$ (standard deviation 2). With a flat prior, what are the mean and variance of the combined estimate?

3.3 Prediction

- Two standardised variables ($\mu_1 = \mu_2 = 0$, $\sigma_1 = \sigma_2 = 1$) have correlation $\rho = 0.8$. Given $x_1 = 2$, find the mean and variance of x_2.

3.4 Conditional covariance

- Using the linear-transformation rule, show that $\boldsymbol{w} = \boldsymbol{z}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{z}_2$ has covariance $\boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$.

3.5 Steady-state Kalman filter

- After many steps the Kalman variance settles to a fixed value v. Write $p = v + q$ and show that $p^2 - q\,p - q\,\sigma^2 = 0$. Find v for $q = 0.1$ and $\sigma^2 = 1$.

3.6 The normal moment generating function

- Show that for $X \sim \mathcal{N}(\mu, \sigma^2)$ the moment generating function is $Z(\ell) = \mathrm{e}^{\mu\ell + \sigma^2\ell^2/2}$, by completing the square in the exponent. What does this say about the cumulants of a normal distribution?

4 Experiments

4.1 Checking the formulae

- The first part checks the conditional formula by keeping samples whose z_2 is close to an observed value; the second runs a one-dimensional Kalman filter

```
import numpy as np

rng = np.random.default_rng(0)

# 1. Check the conditional formula by sampling: keep samples whose z2 is
#    close to an observed value and compare with the formula
mu = np.array([1.0, -1.0, 0.5])
L = rng.standard_normal((3, 3))
Sigma = L @ L.T + 0.5 * np.eye(3)                 # a random covariance
z = rng.multivariate_normal(mu, Sigma, size=2_000_000)
z2_obs = np.array([0.0, 1.0])                     # observe the last two
near = np.all(np.abs(z[:, 1:] - z2_obs) < 0.05, axis=1)
S11, S12, S22 = Sigma[:1, :1], Sigma[:1, 1:], Sigma[1:, 1:]
mean = mu[:1] + S12 @ np.linalg.solve(S22, z2_obs - mu[1:])
var = S11 - S12 @ np.linalg.solve(S22, S12.T)
```

```
print(f"{near.sum()} samples near z2 = {z2_obs}")
print(f"conditional mean: formula {mean[0]:.3f}, samples {z[near, 0].mean():.3f}")
print(f"conditional var:  formula {var[0, 0]:.3f}, samples {z[near, 0].var():.3f}")

# 2. A one-dimensional Kalman filter tracking a drifting position
T, q, sigma2 = 50, 0.1, 1.0
x = np.cumsum(rng.normal(0, np.sqrt(q), T))       # true positions
y = x + rng.normal(0, np.sqrt(sigma2), T)         # noisy observations
m, v = 0.0, 10.0                                  # vague initial belief
est = []
for t in range(T):
    v = v + q                                     # predict
    k = v / (v + sigma2)                          # update
    m, v = m + k * (y[t] - m), (1 - k) * v
    est.append(m)
est = np.array(est)
print(f"RMS error: raw observations {np.sqrt(np.mean((y - x)**2)):.3f}, "
      f"Kalman estimate {np.sqrt(np.mean((est - x)**2)):.3f}, final variance {v:.3f}")
```

- Running it: the sampled conditional mean and variance (1.30, 0.85 from about 900 samples) agree with the formula (1.27, 0.82) to within sampling error; the Kalman filter reduces the RMS error from 0.96 (raw observations) to 0.57, and its variance settles at 0.270, the value found in the exercise

5 Answers

5.1 Fourth moment

- $\mathrm{e}^{\sigma^2 t^2/2} = 1 + \frac{\sigma^2 t^2}{2} + \frac{\sigma^4 t^4}{8} + \cdots$. The coefficient of t^4 is $\mathbb{E}\left[X^4\right]/4!$, so $\mathbb{E}\left[X^4\right] = 24\,\sigma^4/8 = 3\sigma^4$

5.2 Combining sensors

- Precisions 1 and $1/4$, so the mean is $(10 + 12/4)/(1 + 1/4) = 13/1.25 = 10.4$ and the variance is $1/1.25 = 0.8$: closer to, and better than, the better sensor

5.3 Prediction

- $\mathbb{E}[x_2 \mid x_1] = 0.8 \times 2 = 1.6$ and $\mathrm{Var}(x_2 \mid x_1) = 1 - 0.64 = 0.36$

5.4 Conditional covariance

- $\boldsymbol{w} = \mathbf{W}\boldsymbol{z}$ with $\mathbf{W} = (\mathbf{I},\ -\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1})$, so its covariance is

$$\begin{aligned}\mathbf{W}\boldsymbol{\Sigma}\mathbf{W}^{\mathsf{T}} &= \boldsymbol{\Sigma}_{11} - 2\,\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21} + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{22}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21} \\ &= \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\end{aligned}$$

5.5 Steady-state Kalman filter

- At the fixed point $v = (1-k)p = p\,\sigma^2/(p+\sigma^2)$ with $v = p - q$. So $(p-q)(p+\sigma^2) = p\,\sigma^2$, i.e. $p^2 - q\,p - q\,\sigma^2 = 0$
- With $q = 0.1$, $\sigma^2 = 1$: $p = (0.1 + \sqrt{0.01 + 0.4})/2 = 0.370$, so $v = 0.270$ (as in the experiment)

5.6 The normal moment generating function

- $$Z(\ell) = \frac{1}{\sqrt{2\pi}\,\sigma} \int \mathrm{e}^{\ell x - (x-\mu)^2/(2\sigma^2)}\,\mathrm{d}x$$
- Completing the square,

$$\ell\, x - \frac{(x-\mu)^2}{2\sigma^2} = -\frac{\big(x - \mu - \sigma^2\ell\big)^2}{2\sigma^2} + \mu\,\ell + \frac{\sigma^2\ell^2}{2}$$

- The remaining integral is a normal density with mean $\mu + \sigma^2\ell$, which integrates to 1, so $Z(\ell) = \mathrm{e}^{\mu\ell + \sigma^2\ell^2/2}$
- Hence $G(\ell) = \log Z(\ell) = \mu\,\ell + \frac{1}{2}\sigma^2\ell^2$ exactly: $\kappa_1 = \mu$, $\kappa_2 = \sigma^2$, and every higher cumulant is zero. The normal is the distribution whose cumulants stop after the second